EXPERIMENT - 15

15. Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function

Aim:

To read an image in Python and detect its corners using the Harris Corner Detection function in OpenCV.

Procedure:

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Convert the grayscale image to a floating-point format.
5. Apply the Harris Corner Detection function using cv2.cornerHarris().
6. Mark the detected corners in red.
7. Display the original and corner-detected images.

Program:

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    gray = np.float32(gray)

    corners = cv2.cornerHarris(gray, 2, 3, 0.04)
    img[corners > 0.01 * corners.max()] = [0, 0, 255]
```

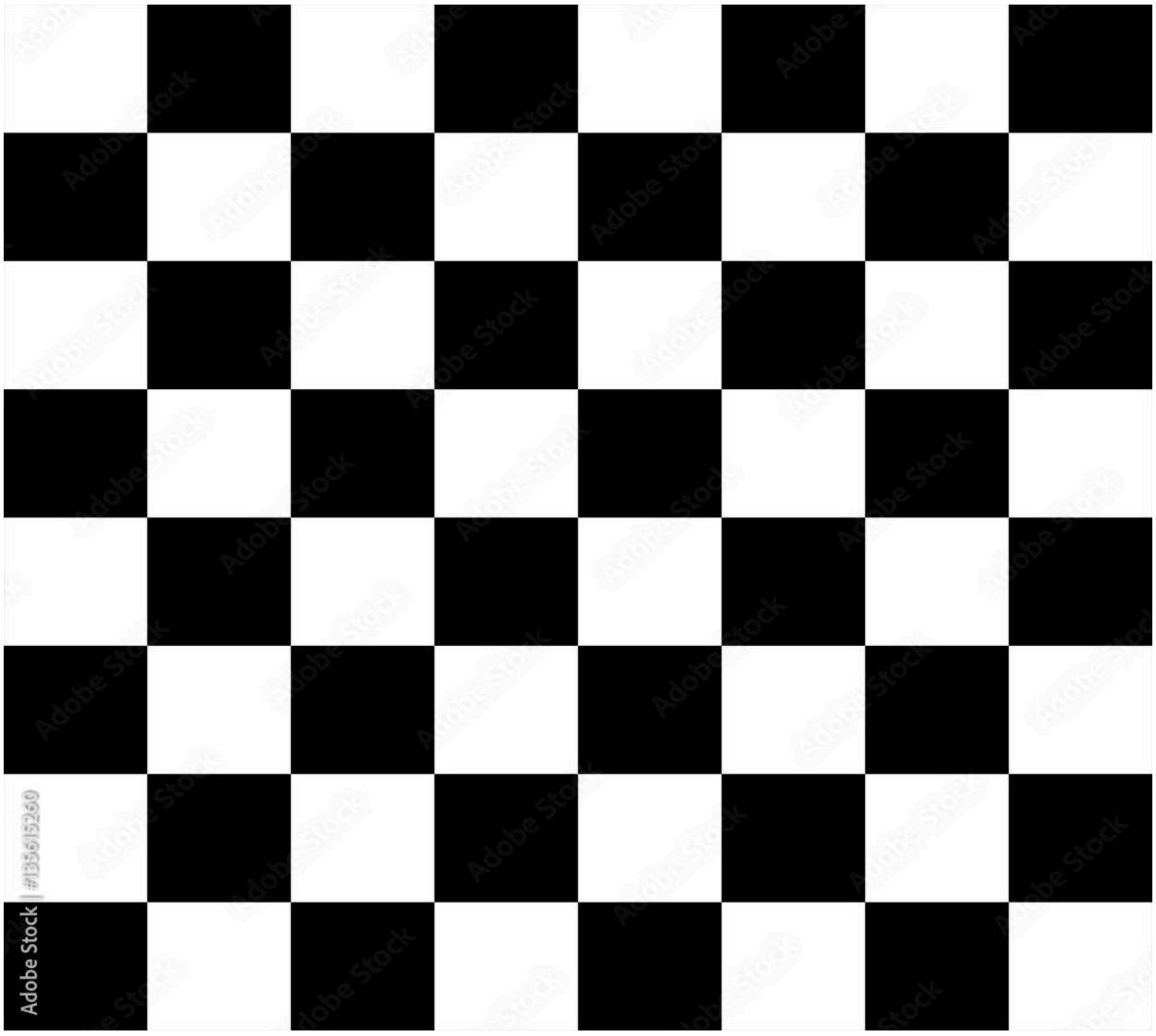
```
original = cv2.imread("image.jpg")

cv2.imshow("Original Image", original)
cv2.imshow("Harris Corner Detection", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

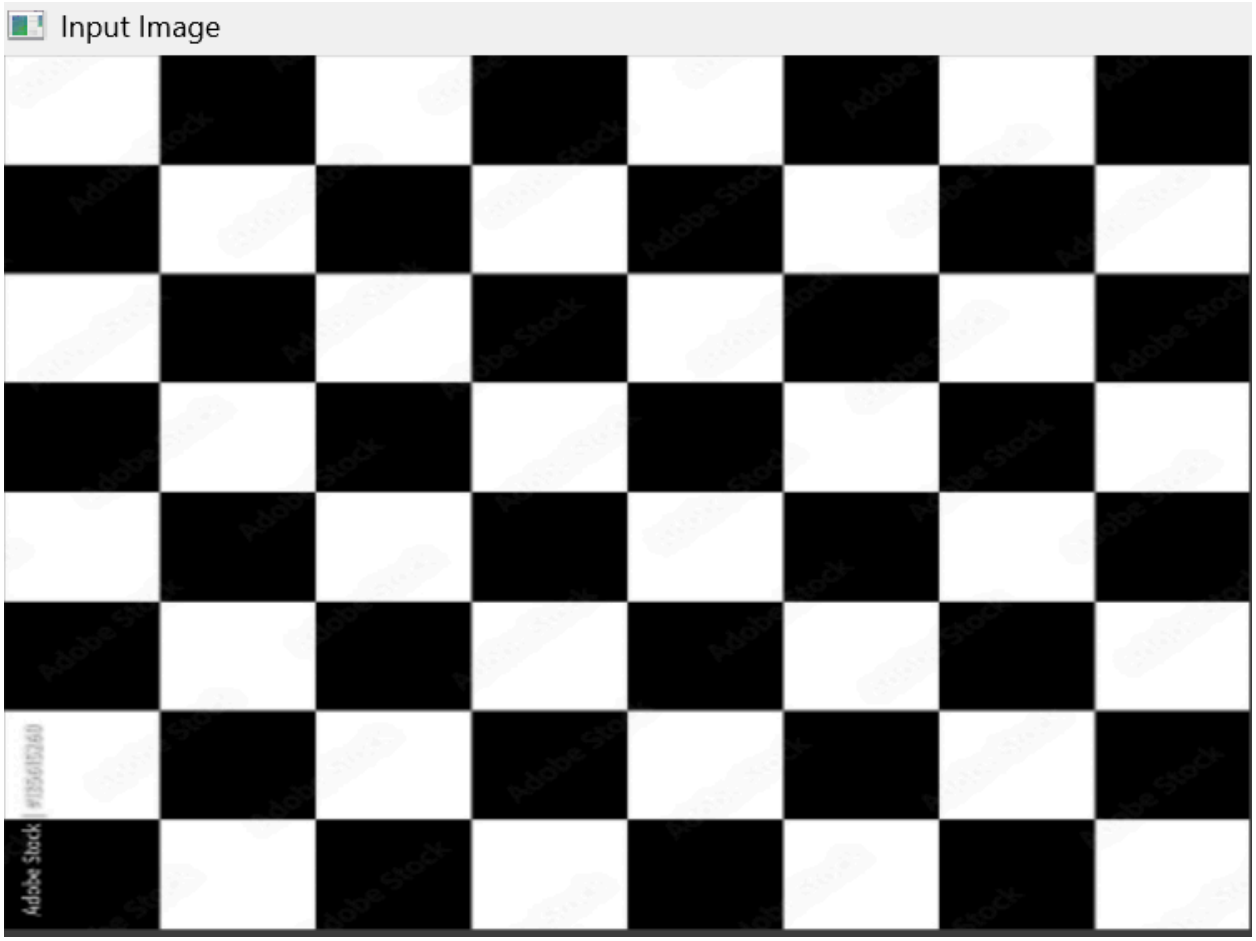
Sample Input:

Input image: image.jpg

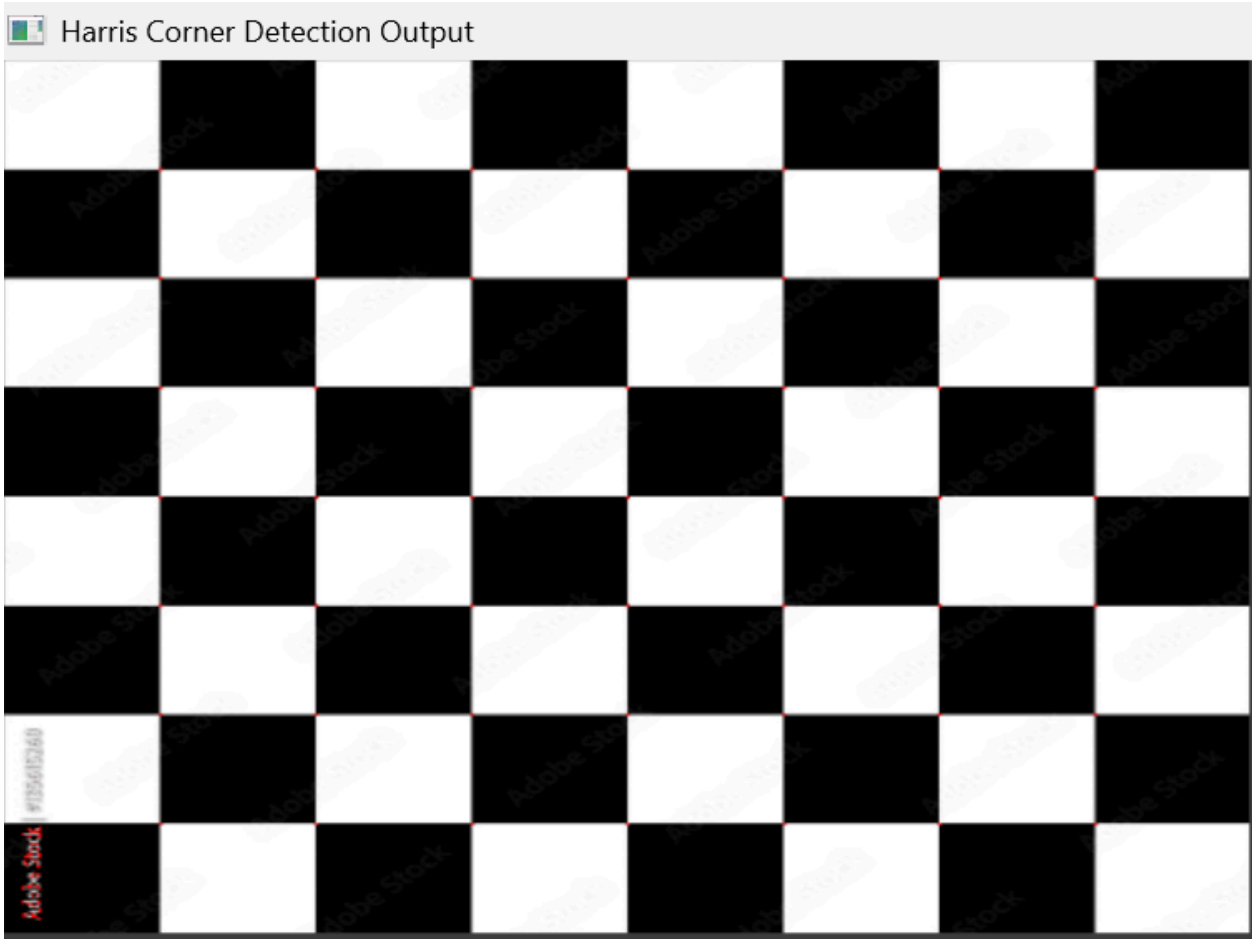


Sample Output:

Original Image



Detected Corners:



Result

Thus, Harris Corner Detection was successfully implemented using Python and OpenCV to identify corners in the given image.